

Transformer Memory for Interactive Visual Navigation in Cluttered Environments

Weiyuan Li , Ruoxin Hong, Jiwei Shen , Liang Yuan, and Yue Lu , *Senior Member, IEEE*

Abstract—Substantial progress has been achieved in embodied visual navigation based on reinforcement learning (RL). These studies presume that the environment is stationary where all the obstacles are static. However, in real cluttered scenes, interactable objects (e.g. shoes and boxes) blocking the way of robots makes the environment non-stationary. Accordingly, the ego-centric visual agent will easily get stuck in the dilemma of finding the next waypoint as it struggles to decide whether to push the obstacles ahead. To handle the predicament, we formulate this interactive visual navigation as Partial Observed Markov Decision Process (POMDP). As the transformer encoder has demonstrated its superior ability to capture the spatial-temporal dependencies in natural language processing. We propose a transformer-based memory to empower the agents utilizing the historical interactive information. However, directly leveraging the transformer architecture in the RL settings is highly unstable. We further propose a surrogate objective to predict the next waypoint as the auxiliary task, which facilitates the representation learning and bootstraps the RL. We demonstrate our method in the iGibson environment and experimental results show a significant improvement over the interactive Gibson benchmark and the related recurrent RL policy both in the validation seen scenes and the test unseen scenes.

Index Terms—Vision-based navigation, reinforcement learning, representation learning.

I. INTRODUCTION

TRADITIONAL navigation tasks require the agent to reach the destination under the premise of avoiding obstacles in a static environment which is applicable in some empty scenarios such as outdoors or in factories. However, the dynamic scenarios inherent to real human environments, such as offices and coffee shops, contain a large number of interactive objects, such as furniture, toys, shoes, etc. shown in Fig. 1. The additional small objects are placed near the robot's shortest path to the goal,

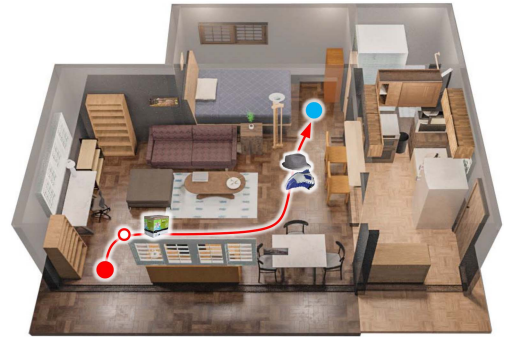


Fig. 1. Interactive Navigation. The interactable objects are placed near the shortest path of the robot. Hence, learning an effective policy to interact with these objects during navigation is vital for the robots.

while the robot has to interact with the environment by pushing or moving the obstacles away to clear the path. Learning to navigate in the interactive environment considering the physical interaction remains several challenges.

In the interactive navigation task, the main challenge is the partial observation problem that arises from the non-stationary environment where the interactable objects are randomly placed blocking the way to the destination. Therefore, the robot has to figure out whether it can push the obstacles out of the way by leveraging the history information. After formulating the task as the POMDP, it is popular to use recurrent neural networks (RNNs) such as LSTM or GRU [1] to aggregate the past observation and actions. However, RNNs' inability to capture the long-term dependencies of the memory is not suitable for this task. We apply a transformer encoder, which is improved from the scene memory transformer [2] with residual connection and local attention mechanism, to handle the long sequence and ensure the stability of training simultaneously.

Generally, the sampling efficiency of the reinforcement learning (RL) from images is rather low, especially in the navigation area where the reward to the robot is sparse. Adopting the additional self-supervised task such as depth prediction [3] in the learning can accelerate the training. The current auxiliary task in navigation mainly focused on helping the robot to perceive the 3D structure of the environment while neglecting the temporal information. Accordingly, we propose to optimize the angle prediction of the optimal waypoint during the exploration, which leverages the spatiotemporal information in the memory. Experiments demonstrate that our auxiliary task facilitates representation learning and bootstraps the RL.

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Weiyuan Li, Ruoxin Hong, Jiwei Shen, and Yue Lu are with the Shanghai Key Laboratory of Multidimensional Information Processing, School of Communication and Electronic Engineering, East China Normal University, Shanghai 200050, China (e-mail: 51205904034@stu.ecnu.edu.cn; iruoxin@163.com; sjwcee@gmail.com; ylu@cs.ecnu.edu.cn).

Liang Yuan is with the Beijing Advanced Innovation Center for Soft Matter Science and Engineering, Beijing University of Chemical Technology, Chaoyang 100013, China (e-mail: yuanliang@mail.buct.edu.cn).

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To tackle interactive visual navigation, we propose a deep reinforcement learning model with transformer-based memory, which can learn to navigate in a cluttered room and interact with the obstacles based on the accumulated experience. Besides, a surrogate objective is proposed to optimize the transformer encoder for a stable and sufficient state representation. We train and evaluate the model in the Interactive Gibson Environment [4] that includes 15 fully interactive scenes based on real-world homes and additional small objects from the Google Scanned Objects dataset. Our approach outperforms the Interactive Gibson Benchmark with over 17% improvement in success rate and 9.2% in success weighted by path length (SPL). The main contributions of this work include:

- 1) The Belief state encoder (BSE) module composed of transformer encoder is introduced to better represent the input visual information and captures the historical interactive information, which produces a more informative state for the following decisions.
- 2) A surrogate objective is designed to provide supervised signals for the representation learning which improves the learning efficiency of the transformer encoder.
- 3) We adjust the structure of the scene memory transformer and introduce the local and casual attention mechanism to promote the convergence of the model.

Section II reviews the related work, followed by the introduction of problem formulation, the framework of the system, and the representation study in Section III. Then, the proposed method is validated by experiments in Section IV. Finally, the conclusion and future work are presented in Section V.

II. RELATED WORK

A. Visual Navigation

Recently, reinforcement learning-based algorithms have been achieving impressive results in visual navigation. In these works, lots of attention are paid to point navigation [5], [6], object navigation [7], [8] and vision-and-language navigation [9], [10]. However, there is a significant performance gap between the navigation performance achieved in the static case versus the dynamic environments with interactable objects. Xia et al. [11] proposed a benchmark for Interactive Navigation in cluttered environments, where the robot is allowed physical interaction with indoor objects (e.g. shoes and boxes). However, their agents make decisions simply based on the current observation O_t , which neglects the historical interactive information. With the transformer-based memory, our approach shows significant improvements over the benchmark [11] and related recurrent RL-based policy [12] in the Interactive Visual Navigation.

B. Memory

In cluttered indoor environments, instead of using LiDAR [13] or externally placed sensors such as ultra-wideband modules [14], learning navigation policy directly from egocentric observations is suffered from the partially observed problem due to the limited receptive field. Besides, the non-stationary environment arising from the interactable objects exacerbates

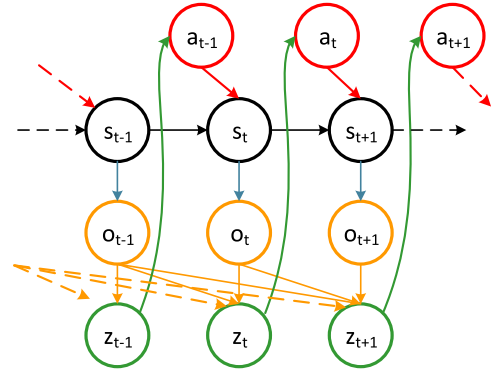


Fig. 2. Probabilistic graphical model: sample observation O_t from original states s_t , estimate representation z_t from observation O_t and learn a policy from representation z_t .

the problem. It is common to aggregate the observations by leveraging implicit memory such as recurrent neural networks (RNNs) [3], [12], while it has optimization problems over the long sequence in the navigation. Besides, others explicitly exploit the memory by the map-based memory [15], [16], which is added hand-crafted feature or implemented with complex systems. Furthermore, the attention mechanism [17] is introduced to the visual navigation [2], which can exploit spatio-temporal dependencies in the aggregated observations. However, training such transformer-based RL is highly unstable. We adjust the structure of the transformer [17] and introduce the local attention mechanism to promote the convergence of the model.

C. Representation Learning

Learning the auxiliary tasks is considered an effective method to improve the performance or the sample efficiency of the reinforcement learning [18]. In the visual navigation task, RL-based algorithms often construct self-supervised auxiliary tasks such as predicting the external environment [19] or the pose estimation of the robot itself [6]. Mirowski et al. [3] attempted to learn an intrinsic representation of space, geometry, and movement by depth prediction and loop closure prediction. [20], [21] proposed to learn a sub-goal of navigation which is setting the current waypoints of the robot. Inspired by their works, we leverage the optimal waypoint calculated in the training scenes as the supervised signal. With this auxiliary task, we can precisely predict the angle of the optimal waypoint and train the transformer-based RL algorithm effectively.

III. APPROACH

Problem Formulation: In our analysis, as shown in Fig. 2, the original state S is unavailable for the agent which contains all the objective information from the environment, including the global position of the robot, the map of the environment and whether the objects are interactable. We assume that the original state S is Markovian. Nevertheless, the sampled observation O from the state space S is partially observed as the ego-centric observed robot in the cluttered environment can only perceive

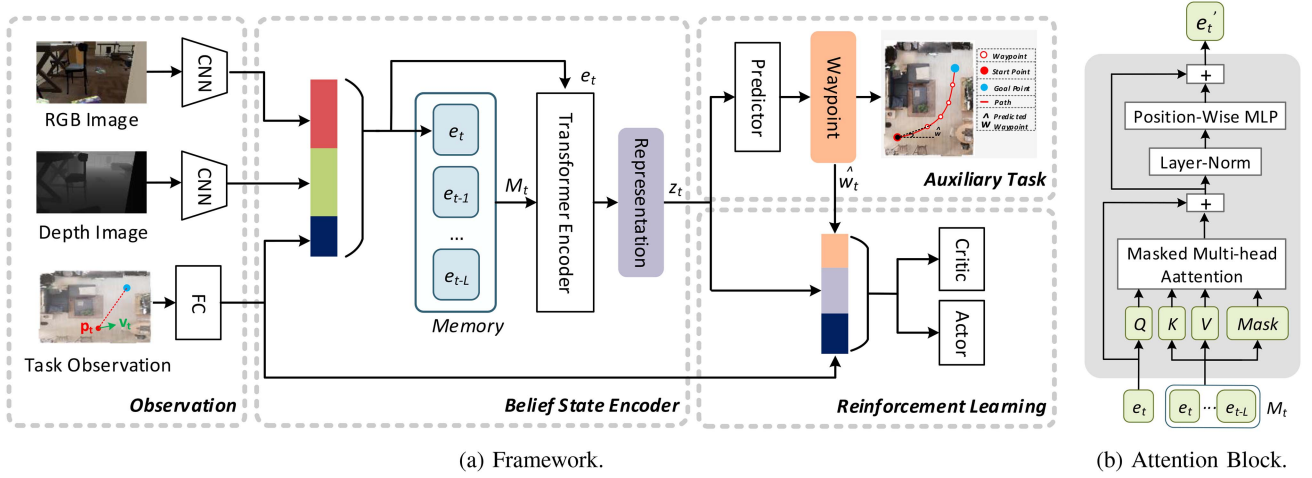


Fig. 3. The schematic overview of the system. (a) The framework of our model. The model includes four modules: an Observation module, a Belief State Encoder, a State Representation module, and a Reinforcement Learning module. (b) The attention block. The transformer encoder in (a) is composed of P stacked attention block.

the obstacles and routes ahead. To alleviate the problem of partial observability, we define a state representation ϕ_Z as a stochastic mapping from the historical observation sequences to a representation space Z : $p(Z = z_t | O_0, O_1, \dots, O_t)$. Accordingly, we formulate the interactive navigation task as a partially observed Markov decision process. Formally, the POMDP is defined as a tuple (Z, A, τ, r, γ) , where Z, A, r are the belief state, action, and reward. τ is the belief state transition function and $\gamma \in [0, 1]$ is the discount factor.

Fig. 3 provides the schematic overview of the system. We leverage the transformer architecture to produce the belief state Z . Instead of training the transformer encoder end-to-end, we propose a surrogate objective to optimize the state representation. Based on Z with sufficient information, the robot can learn an effective and robust policy to navigate in these cluttered environments.

A. Observation Encoder

At each time step t , the robot receives an observation sampled from the environments $O_t = (I_t, p_t, v_t)$, where I_t is the ego-centric visual observation consisting of RGB and depth images; p_t denotes the goal position relative to the robot in polar coordinates and v_t represents the current linear and angular velocities.

As shown in Fig. 3(a), these ego-centric observations are encoded with two convolutional neural networks (CNN) and a fully-connected network (FC) which respectively constitute an embedded feature for the later proceedings.

$$e_t^o = \text{Cat} \left(\psi_{\text{cnn}_1} (I_t^{\text{RGB}}), \psi_{\text{cnn}_2} (I_t^{\text{depth}}), \psi_{\text{fc}} (p_t, v_t) \right). \quad (1)$$

B. Belief State Encoder

Instead of taking the current embedding as the state, we use the embedding as the query to extract useful information from the memory. So the state encoder exploits the transformer architecture to handle the historical observations sequence.

Memory: The memory is initialized as an empty set at the beginning of each episode. During the robot exploration period, we maintain the M_t in the fixed-length l by storing the embedding of the current observation O_t .

$$M_t = \begin{cases} M_{t-1} & \cup \{e_t\}, \quad t < l \\ M_{t-1}[1:l] & \cup \{e_t\}, \quad t \geq l \end{cases} \quad (2)$$

Transformer Encoder: Masked attention mechanism is detailed in [17], it can be formulated as a function Att that takes query $Q \in \mathbb{R}^{s_1 \times d_k}$, key $K \in \mathbb{R}^{s_2 \times d_k}$, value $V \in \mathbb{R}^{s_2 \times d_v}$ and mask $m \in \mathbb{R}^{s_1 \times s_2}$ as the input, where d_k is the dimension of the query, d_v is the dimension of the value and m is the matrix added to the dot product of the queries and keys. The output of the Att has s_1 elements of d_v .

$$\text{Att}(Q, K, V, m) = (\text{softmax}(m + (QK^T))) V. \quad (3)$$

Motivated by GTrXL [22], based on the above Att function, we modify the position of the layer normalization (LN) to reconstruct the AttBlock . As shown in Fig. 3(b), the AttBlock takes the $e_t \in \mathbb{R}^{1 \times d_e}$, $M_t \in \mathbb{R}^{l \times d_e}$ as the input, where l is the size of memory, d_e is the dimension of the embedding. Therefore, the embedding of the current observation can directly flow through the AttBlock without any transformation, which stabilizes the transformer-based RL algorithm substantially.

$$\text{AttBlock}(e_t, M_t) = \text{FC}(\text{LN}(H)) + H,$$

$$\text{where } H = \text{Att}(e_t W^Q, M_t W^K, M_t W^V, \text{Mask}(M_t)) + e_t. \quad (4)$$

where the Mask is the function calculated according to the dimension of M_t . $W^Q \in \mathbb{R}^{d_e \times d_k}$, $W^K \in \mathbb{R}^{d_e \times d_k}$ and $W^V \in \mathbb{R}^{d_e \times d_v}$ project the e_t and M_t to Q, K and V respectively. The d_v and d_k are set equal to d_e considering the residual connection of the current observation embedding e_t . Accordingly, the output of the AttBlock has 1 element of d_e , the same shape as that of input e_t .

The transformer encoder comprises a stack of P AttBlock, where the output of each AttBlock can be viewed as the query over the next AttBlock.

$$\text{Encoder}(e_t, M_t) = \text{AttBlock}(\text{AttBlock}(e_t, M_t), M_t). \quad (5)$$

The residual connection of the network structure above and the following proposed supervised signal reduce the training complexity of the transformer network.

C. Learning Objective of the State Representation

As shown in Fig. 3(a), the optimal shortest path between the agent and the goal is discretized into several waypoints. Intuitively, if the state representation ϕ_Z is sufficient for the optimal navigation policy $\pi^*(a_t|z_t)$, it should preserve enough information for the prediction of the next waypoint. Accordingly, the prediction of the next waypoint can be regarded as the surrogate objective to optimize the state representation ϕ_Z . Since the state of the waypoints prediction contains sufficient mutual information with the representation of the navigation.

During the training stage, the waypoints sampled from the shortest path can be calculated by A^* algorithm on the global map information S . Therefore, the surrogate objective is to minimize the KL divergence between the first optimal waypoint w given the state s_t and the probability distribution of the output $\hat{w}$ of the prediction network given z_t .

$$\begin{aligned} \text{KL}(p(w|s_t)||p(\hat{w}|z_t)) &= \mathbb{E}_{p(w|s_t)}[\log p(w|s_t) - \log p(\hat{w}|z_t)] \\ &= \mathcal{H}(w) - \mathbb{E}_{p(w|s_t)}[\log p(\hat{w}|z_t)], \end{aligned} \quad (6)$$

where $\mathcal{H}(w)$ is the entropy of the optimal waypoint and it does not influence the optimization process. Then, the optimization objective of the state representation is to maximize $\mathbb{E}_{p(w|s_t)}[\log p(\hat{w}|z_t)]$.

Accordingly, the loss function of the surrogate objective to predict the next optimal waypoint contains two components specifying: 1) the angle prediction of the waypoint $\mathcal{L}_\theta$, and 2) the distance prediction of the waypoint $\mathcal{L}_d$.

$$\mathcal{L} = \sum_{i=1}^L \left(1 - \cos(\theta_i - \hat{\theta}_i)\right) + \sum_{i=1}^L (d_i - \hat{d}_i)^2, \quad (7)$$

where L is the length of the sequence, θ and d are the angle and distance of the optimal waypoint relative to the robot respectively. $\hat{\theta}$ and $\hat{d}$ are the corresponding estimation calculated by the predictor. With the help of waypoint prediction, we can train the transformer-based model end to end without any pre-trained visual encoder for the scene memory transformer [2], [23].

D. Reinforcement Learning Policy

The policy π of the interactive navigation task is trained based on the actor-critic framework. It maps the belief state z_t with residual task observations (p_t, v_t) and predicted waypoints $\hat{w}_t$ to a continuous action a_t for interacting with the environment. The optimization goal is finding a policy that maximizes the cumulative discounted return of the robot.

State: The state representation Z , which is the stochastic mapping from the historical observation sequences: $p(Z =$

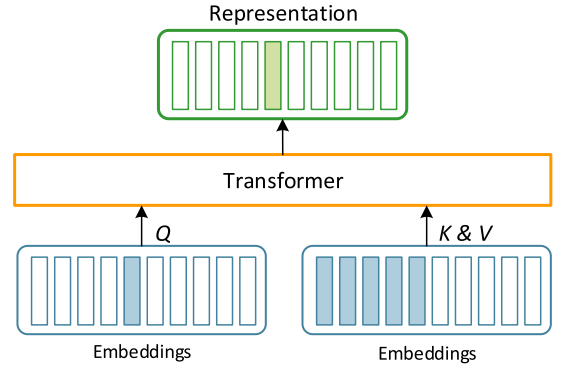


Fig. 4. The diagram of how to calculate the representations. The memory that contains a sequence of embeddings (blue boxes) is taken as the input of the transformer. Under the causal mask (shaded areas), the embedding of the current observation is selected as the query and the embeddings of the historical observations are the keys and values. The output of the transformer encoder is the representation z_t .

$z_t|O_0, \dots, O_t)$, is calculated according to the Fig. 4. Besides, the embeddings of the predictor $\hat{w}$ and the current task observation (p_t, v_t) are also fed into agents.

Action: The action space $\mathcal{A} \in \mathbb{R}^2$ of the robot is the desired normalized continuous linear and angular velocity. We formulate the policy as the $\pi_{a_t \sim \mathcal{A}, z_t \sim \mathcal{Z}}(a_t|z_t)$.

Reward: The reward consists of sparse success reward, potential reward, and interaction reward. The potential reward is based on the geodesic distance between the agent and the goal considered a reward shaping. The interaction reward is calculated from the force applied to the interactable objects, which is taken as the penalty for interacting with the environments. In general, the reward is designed to guide the agent to navigate in cluttered environments efficiently while preserving energy.

E. Training

The detail of the training algorithm is summarized in Algorithm 1. We implement distributed soft actor-critic algorithm based on vanilla SAC [24], and the model is updated by the $\mathcal{L}_{sac}$ and surrogate objective simultaneously. Additionally, the data augmentation techniques(e.g. random crop) are utilized to improve sample efficiency of the reinforcement learning as detailed in data regularized Q [25].

Implementation Details: During the training stage, we sample a batch N composed of a sequence of length L from the replay buffer, where N is 32 and L is the max timesteps of the episode 500. They are fed into the observation encoder where each CNNs block has 6 convolutional layers. The output of the observation encoder is concatenated into a batch of embeddings $E_b \in \mathbb{R}^{N \times L \times d_e}$. The following transformer encoder stacks 2 attention blocks and each block uses multi-head attention mechanisms [17] with 4 heads. The dimension of keys d_k and values d_v are both 256. The predictor is composed of 2 fully connected layers. Besides, all the fully connected layers in the attention blocks are 256 dimensions and ReLU is used as the non-linear activation function.

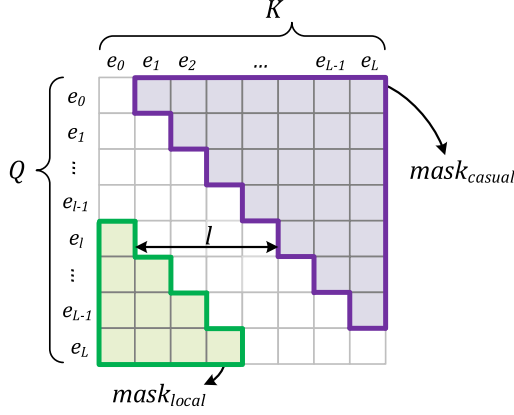


Fig. 5. Mask in the attention block.

Attention Mask: For parallel computing, we propose the attention mask $\text{att_mask} \in \mathbb{R}^{L \times L}$ which is applied to the transformer encoder to ensure the consistency of the results calculated during the training and exploration stage. With the attention mask, the transformer encoder is formulated as $\text{Encoder}(E_b, E_b, \text{att_mask})$, where the attention mask is the summation of the casual mask $\text{mask}_{\text{casual}}$ and the local mask $\text{mask}_{\text{local}}$. As shown in Fig. 5, the shaded area is filled with $-\text{inf}$, and the length of memory is l . The casual mask $\text{mask}_{\text{casual}} \in \mathbb{R}^{L \times L}$ is an upper triangular matrix which prevents e_i , where $i \in (0, L)$, from using the information embedded in the e_j , where $j \in (i, L)$, in the calculation of the attention. The casual mask ensures the causality of training. The Local mask $\text{mask}_{\text{local}} \in \mathbb{R}^{L \times L}$ is a lower triangular matrix shifted down by l rows which prevents e_i , where $i \in (0, L)$, from using the information embedded in the e_j , where $j \in (0, i - l)$, in the calculation of the attention. The local mask ensures that attention is attracted to the same area of training as that of exploration.

IV. EXPERIMENT

To analyze the effectiveness of our proposed method, we present experiments studying representation learning that are optimized by the estimation of the distance and angle of the next waypoint. We also include the performance of random walk, Xia et al. [11], Savva et al. [12] and Fang et al. [2] to further evaluate the proposed method.

A. Experiment Setup

Environment and Datasets: The experiment is conducted on the Interactive Gibson Environment [11], where the robots are allowed (or even encouraged) to interact physically with objects. The extra objects (e.g. shoes and toys) from the Google Scanned Objects dataset are also added to the environment to simulate real-world clutter. The Interactive Gibson Environment retains the photorealism and the semantic complexity of the original Gibson Navigation environment which includes over 1400 floor spaces from 572 full buildings. Besides, 1984 interactable CAD model alignments of furniture are included to augment 106 scenes. In each episode, the environment, the location to place

Algorithm 1: Training Algorithm.

```

1 BSE  $\leftarrow$  initialize belief state encoder
2 OE  $\leftarrow$  initialize observation encoder
3  $p \leftarrow$  initialize predictor
4  $\pi \leftarrow$  initialize policy
5  $Q \leftarrow$  initialize Q value function
6 while not done do
7    $\mathcal{B} \leftarrow \emptyset$  initialize replay buffer
8   for trajectory  $i = 1, \dots, m$  do
9      $\tau^i \leftarrow \{o_0, a_0, o_1, \dots, o_T\}$  collect trajectory with
        $\pi$ 
10    for time step  $t = 0, \dots, T - 1$  do
11      calculate optimal next waypoint  $w_t$  according
        to global map information
12      calculate rewards  $r_t$ 
13      record  $w_t, r_t$  in  $\tau^i$ 
14    end for
15    store  $\tau^i$  in  $\mathcal{B}$ 
16  end for
17
18  Update encoder:
19  for update step  $= 1, \dots, n$  do
20     $b^\pi \leftarrow$  sample batch of  $K$  transitions
       $\{(o_j, o'_j, w_j, w'_j)\}_{j=1}^K$  from  $\mathcal{B}$ 
21    update BSE, OE and  $p$  according to Equation 6
      and Equation 7 using  $b^\pi$ 
22  end for
23
24  Update actor-critic:
25  for update step  $= 1, \dots, n$  do
26     $b^\pi \leftarrow$  sample batch of  $K$  transitions
       $\{(o_j, o'_j)\}_{j=1}^K$ 
27     $b^{\hat{w}} \leftarrow$  predict the batch of next waypoints given
       $b^\pi$ 
28     $b^z \leftarrow$  calculate the batch of representations given
       $b^\pi$ 
29     $b^s \leftarrow (b^z, b^{\hat{w}})$  formulate the state representation
      for agent
30    update  $Q$  and  $\pi$  respectively according to data
      regularized  $Q$  [25] using  $b^s$ 
31  end for
32 end while

```

the additional objects, the starting location, and the goal location are randomly sampled. The starting and goal locations are at a distance ranging from 3 to 10 meters. The agent is required to reach the goal according to the given information, including RGB+D images, goal position relative to the robot in polar coordinates, and the current linear and angular velocities. The episode terminates when the agent reaches the goal location or the number of steps taken by the agent exceeds 500. We randomly chose 5 scenes from 8 fully interactive scenes as the training environments and others are tested as unseen environments. With asynchronous sampling in 5 training environments with 5 different random seeds, the training process is performed via 4 GPU NVIDIA RTX 2080 Ti for about 10 hours.

Evaluation Metrics: We use success rate (SR), success weighted by path length (SPL), effort efficiency (EE), and interactive navigation score (INS) to evaluate the model [11].

- 1) SR is the ratio of successful episodes to all episodes.
- 2) SPL is the standard metric evaluating successes according to their adherence to the shortest path [26]. The summary

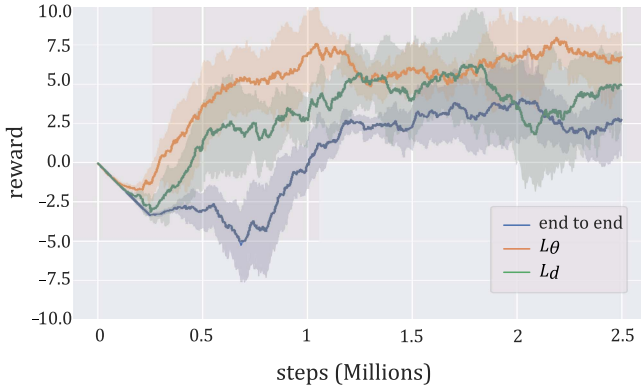


Fig. 6. Performance of RL agents trained with different state representations, which are optimized with the corresponding surrogate objective.

measure across the test is as follows:

$$SPL = \frac{1}{N} \sum_{i=1}^N S_i \frac{l_i}{\max(p_i, l_i)}, \quad (8)$$

where N stands for the number of conducted episodes. l_i is the shortest path distance of the agent from the starting point to the goal in episode i , and p_i represents the distance actually taken by the agent. S_i indicates the success in episode i using the binary code.

- 3) EE describes the efficiency of the agent's efforts to achieve the goal, including moving its own body and interacting with the objects in the environment [11]:

$$EE = 0.5 \left(\frac{m_0 l_0}{\sum_{i=1}^K m_i l_i} + \frac{TG}{TG + \sum_{t=0}^T F_t} \right), \quad (9)$$

where m_i represents the masses of the robot ($i=0$) and the objects. We denote l_i as the distance taken by i th object. The gravity force of the robot G is calculated by $m_0 g$ and F_t stands for the amount of force the robot applied on the environment at time t where t is from 0 to T .

- 4) INS captures both aspects of SPL and EE, denoted as:

$$INS_\alpha = \alpha SPL + (1 - \alpha) EE. \quad (10)$$

The importance of the trade-off between SPL and EE is controlled by α .

B. Representation Study

Optimizing the estimation of the angle between the agent and the next waypoint successfully represents the original observations and it reaches the highest reward as shown in Fig. 6. As the waypoints are sampled at 0.5 m intervals on the ground truth path, the offset within 0.5 m can be tolerated. Taking the offset into consideration, the deviation of the angle between the agent and the closest waypoints is negligible. Hence, the estimation of the angle can produce effective information.

Learning a representation by optimizing the estimation of the distance between the agent and the next waypoint may have difficulty representing the optimal policy. If the agent's forward direction does not change, then any distance estimate on this

path is valid, and it becomes meaningless to consider $\mathcal{L}_d$ in this case. Indeed, experiments show that learning a policy from the learned representation incurs a lower reward.

As shown in Fig. 6, the end-to-end RL performs poorly and the reward of it remains the lowest during almost the entire training process. Without representation, learning a policy directly from the original observations, the agent failed to capture the information of the next waypoint and have trouble planning the route.

Hence, optimizing the estimation of the angle between the agent and the next waypoint produces a sufficient representation and it's more effective than learning a policy directly from original observations. The following quantitative indicators are computed by leveraging the predicted angle of the waypoints.

C. Performance Comparison

We involve the performance of several baselines including Random, Xia et al., Savva et al. and Fang et al. [2]. Random policy uniformly samples from the continuous action space. Interactive Gibson Benchmark proposed by Xia et al. [11] evaluates the performance of a series of algorithms based on reinforcement learning in the Interactive Gibson Environment. Among them, the vanilla SAC-based algorithm obtains the best scores. We also analyze the performance of an RL Policy [12] with a convolutional network followed by a GRU, as it is considered the most common memory-based method. Fang et al.'s work [2] first utilized the transformer-based memory to perform long-horizon tasks in a partially observable environment. However, the attention mechanism in the temporal dimension of the transformer brings stronger fitting ability and is also more difficult to converge. The results are shown in Table I. The SR of random indicates the complexity of the task, randomly sampling the policy from the action space is impossible to work. Notably, our method outperforms the baselines in SR, SPL, and INS while the EE is slightly lower than Savva et al.'s work, which reveals that our method increases the interaction with objects and achieves better results. Particularly, the success rate increases by over 10% compared to the best of the baseline. The comparison with baselines justifies the effectiveness of our method.

D. Ablation Study

We present the additional ablation study to better understand the impact of the BSE and the state representation.

Ours w/o $\mathcal{L}_\theta$: As mentioned above, with the help of angle prediction, the model can be trained end to end without the pre-trained network. We remove the auxiliary loss module to provide us with an insight into its importance of it. Intuitively, we employ the pre-trained CNN as the visual encoder which is the same as the baseline. The parameters of the embedding network are frozen for other parts of the model.

Ours w/o BSE: For the interactive navigation, the output of the transformer encoder is a part of the belief state which helps the agent to grow the knowledge of the environment. We remove the BSE module and retain the other parts of the structure. Without the timing information, for a fair comparison, the current embedding is converted through several fully connected layers to make

TABLE I
PERFORMANCE COMPARISON. WE SHOW THE RESULT OF THE COMPARISON OF THE PERFORMANCE OF OUR METHOD WITH THE PREVIOUS STATE-OF-THE-ART METHOD ALONG WITH THE ABLATION STUDY

Methods	SR (%)	Validation Seen			SR (%)	Test Unseen		
		SPL	EE	INS _{0.5}		SPL	EE	INS _{0.5}
Baselines:								
Random	1.0	0.4	-	-	0.6	0.2	-	-
Xia et al. [11]	68.8	41.5	92.5	67.0	59.7	37.0	92.3	64.6
Savva et al. [12]	72.3	43.1	95.7	69.4	66.7	40.3	95.6	67.9
Fang et al. [2]	69.4	42.8	93.6	68.2	54.6	35.6	92.2	63.9
ours	82.8	48.7	94.5	71.6	76.7	46.2	93.9	70.0
Ablations:								
ours w/o $\mathcal{L}_\theta$	70.8	43.0	94.4	68.7	61.0	38.8	92.4	65.6
ours w/o BSE	78.2	45.6	93.4	69.5	68.3	41.7	92.7	67.2

In every scenario, we test 100 episodes with randomly generated start and end points to obtain the average. For the indexes shown in the table, the larger the number is, the better the performance. The α of INS is 0.5 as we take the effort efficiency and SPL as equal importance.

The bold value represents the best performance for that metric.

TABLE II
PERFORMANCE COMPARISON AMONG THE DIFFERENT STATE ENCODER STRUCTURES WITH THE AUXILIARY LOSS IN THE TEST UNSEEN SCENARIOS

	SR (%)	SPL	EE	INS _{0.5}
Single + $\mathcal{L}_\theta$	68.3	41.7	92.7	67.2
GRU + $\mathcal{L}_\theta$	70.0	42.4	94.6	68.3
BSE + $\mathcal{L}_\theta$	76.7	46.2	93.9	70.0

sure that the number of parameters is the same as the original transformer encoder. Then the converted current embedding and the predicted waypoints are concatenated as the belief state.

Summary: The performance of ‘ours w/o $\mathcal{L}_\theta$ ’ drops noticeably, which indicates that the angle prediction is essential when the robots planning the routes. For instance, when planning, if the robot can predict the angle of the destination, it is easier to get the right path. Besides, it also demonstrates that the proposed auxiliary tasks facilitate the representation learning of the transformer substantially. The performance of ‘ours w/o BSE’ ablation decreases because the temporal information in the memory is essential for the robot to fully understand the environment and overcome the problem of partial observation.

E. Further Study on State Encoder

The robust representation, optimized by our proposed auxiliary task, is shown to be indispensable in ‘ours w/o $\mathcal{L}_\theta$ ’. To further demonstrate the effectiveness of our architecture changes, we conducted extra experiments on how recurrent RL policy [12] performs with auxiliary tasks. As shown in Table II, the overall performance of recurrent policy is still slightly lower than our method, which is consistent with our insights as the transformer is the more expressive architectural alternative to the recurrent state encoder. Besides, the results of ours w/o BSE demonstrates that the single-frame observations policy [11] plus additional auxiliary task achieves the worst performance among all the state encoder.

V. CONCLUSION

In this letter, we propose transformer-based reinforcement learning to tackle the interactive visual navigation task. Our

experiments empirically validate the ability of our proposed methods. BSE offers an easy-to-train, but the substantially more expressive architectural alternative to other state encoders in interactive visual navigation. Additionally, a temporal auxiliary task to predict the next waypoint is utilized for the extra gradient information and better state representation that contains valuable historical information. Therefore, the robot can utilize the received observation at each step and grow its knowledge of the interactive environment. Furthermore, the proposed methods can also be generally applicable to point navigation, object navigation, or even vision language navigation.

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